Paper Trail

CTF	Angstrom 2019
Category	misc
Difficulty	Easy
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Description

> Something is suspicious about defund's math papers. See if you can find anything in the network packets we've intercepted from his computer.

`address` : https://2019.angstromctf.com/challenges

Introduction

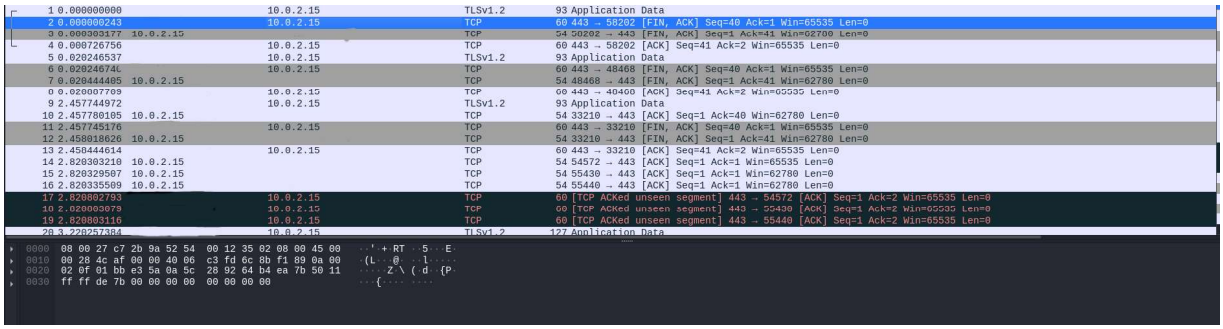
In this challenge we have a `.pcap` file. Let's go to know something about this kind of Files.

A `.pcap` file stores a network traffic over a network inteface (as eth0, wlan0 ans so on).

Mainly this kind of files is created by a software as WireShark or tshark.

On a `.pcap` file we can apply search operations and filters over network packets captured by our software and stored into the file.

The following screenshot shows how a stored packet appears in Wireshark.



Solution

To solve this easy challenge we just need to read the `.pcap` file. If you try to open it in WireShark and read the packets you will see that there are few packets containing just a character and first 5 of them are `a`, `c`, `t`, `f`, `{`.

A speed way to collect every char is using tshark. The following command solves the challenge.

```
┌──(kali㉿kali)-[~/…/AngstromCTF/2019/misc/paperTrail]
└─$ tshark -r  paper_trail.pcapng -Y "irc" -e "irc.response.trailer" -T fields | tr -d  '\n'
I have to confide in someone, even if it's myselfmy publications are all randomly generated :(actf{fake_chat.freenode.netmath_papers}
```

Pay attention because there is some garbage you have to remove `chat.freenode.net`

```
┌──(kali㉿kali)-[~/…/AngstromCTF/2019/misc/paperTrail]
└─$ tshark -r  paper_trail.pcapng -Y "irc" -e "irc.response.trailer" -T fields |grep -v "chat.freenode.net" |tr -d  '\n'
I have to confide in someone, even if it's myselfmy publications are all randomly generated :(actf{fake_math_papers}
```

Let's explain the command:

1. `tshark`
   a. `-r` Opens the file in read mode
   b. `-Y "irc"` Filters just the IRC traffic
   c. `-e "irc.response.trailer` takes just the packet trailer
   d. `-T fields` it's required by the `-e` parameter
2. `grep`
   a. `-v` Inverts the selection and takes all of lines doesn't match "chat.freenode.net"
3. `tr -d "\n"` Removes the new line and prints the output in a single line

Flag : actf{fake_math_papers}

Links and tools